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OpenTakServer Setup:

Hardware Needed:

- Raspberry Pi 5 - https://www.pishop.us/product/raspberry-pi-5-8gb/?src=raspberrypi
- Mini Wireless Router - https://www.amazon.com/dp/B07794JRC5?ref=ppx_yo2ov_dt_b_fed_asin_title
- Battery Bank - https://www.amazon.com/dp/B0DBG2D5DF/ref=sspa_dk_detail_1?psc=1&pd_rd_i=B0DBG2D5DF&pd_rd_w=PCO0N&content-id=amzn1.sym.953c7d66-4120-4d22-a777-f19dbfa69309&pf_rd_p=953c7d66-4120-4d22-a777-f19dbfa69309&pf_rd_r=4XQ7PPW3MET659613SE3&pd_rd_wg=tBAv4&pd_rd_r=e04516f1-c8bd-4a5a-80d1-d65eedb070db&s=wireless&sp_csd=d2lkZ2V0TmFtZT1zcF9kZXRhaWwy
- Meshtastic Device - https://www.amazon.com/ESP32-V3-Module-3000mAh-Battery/dp/B0D7HSHTNX/ref=sr_1_4?crid=1N6MK9LUGZFUP&dib=eyJ2IjoiMSJ9.UfNDfR3gwmCbI-VTkCjLGwxuyQPc4klnnPFjhc0ROue-ZjtfzAWoyliJUVY7psf3NKIGUS7p_RCDhyu1nR3fyv1lhFDqOTPFz_5RQOeV2SgL9yOzX9kFmwAr4SYzA0MVD7XmWC_RIv1kBrYRkuSgu6dZQY_01v8i_VfHsqAZ6FtO6XKqPBcOHvQNyGSFfjPnUkb4KykHLOKqT6x0MBtCZP-a_gCc1pyTfgoXaxrR7_U._IX9qoGut_CrwzO7cW59jYJt4OPDO7kgGVW6ICcluds&dib

_tag=se&keywords=meshtastic&qid=1735356773&sprefix=meshtastic%2Caps%2C115&sr=8-4&th=1

Optional Hardware Parts:

- GPS Module - https://www.amazon.com/Navigation-Positioning-Microcontroller-Compatible-Sensitivity/dp/B0B31NRSD2?dib=eyJ2IjoiMSJ9.NqJbTduGlX7ig1z3YrsPoXS990XIHMs0rKdCzj0QIlPIEvobIz5TD2NSTedRGUnKVU5-BwmU4yzg6_Skr193r-IrzvTkz_x1c7fSyOYfivHV2ylcqozpaND5JMhZxoYsacWWIIulwQGKgftnRt71K1tUsu-EGWqdL9pJum_k7t5WTtbB1wUzvVwBi2FzEibAhGfN5mfAXLp3yg3YFj_N8oDJbWj3FYE288yCMiD68Uw.m0Y-D5ZnDBpY87MmY_mAAWuJ60tMrRbqHxTl4r5qpCw&dib_tag=se&keywords=Raspberry+Pi+GPS&qid=1734724718&sr=8-3
- 5 inch touchscreen - https://www.amazon.com/Hosyond-Raspberry-1024×600-Capacitive-Compatible/dp/B0BKGCB18T?crid=36IN9514D44SU&dib=eyJ2IjoiMSJ9.NwVrLcAK31oKp1_gEHkWmTOXB72WRklu_UTaLNUnAU6wZtABUkafFerp0ySFB2aGqupEAAGfvVTvGfaTqGTpnYauyFTQNXgjf3sWBHkuYIjssK5B18vSYt4mQUDPUjlhEJWflE7qP_3mxt2Fn3WQ4urcBXwGgXJG8ovyJiGYCw-6JQwB5h6oFRUL1sZJxkXfS7rd38obkucxeWDQxTylUYanXt-xtL_54wy-FdboNww.29WAIfz9FOgLAgHsyhNhZhNT5e0gbStnmgVdTGocamA&dib_tag=se&keywords=7in%2Btouchscreen%2Bfor%2Braspberry%2Bpi&qid=1735000689&sprefix=7in%2Btou%2Caps%2C138&sr=8-6&th=1
- Keyboard with Touchpad - https://www.amazon.com/Vilros-Inch-Keyboard-Touchpad-Great-Raspberry/dp/B08PW3LR3W?crid=2CG5FA6E75KTV&dib=eyJ2IjoiMSJ9.eRLfEXCz5hs18diT-l0VeNtvWFCKgvAkxBtDr9NbIFUPjVdAnQKp_hSBNc3Xd1UIZO5JAKSzme29hOogDT2OTqYfgt9IiikEz_BhnpVEpefxluqwurVE-2qla-QuPgzA92YA7CtBaElymylMtPhiSrVCi8ZPwTkxGAxnJGCfBalPDtISv2GRe4bwv-ljOFXlV1ozKTa7_APCnaq00vh9WeWsRQ46E5xXN5vekl7EMVY.0x80N6ShaNuSavg75cyv0OXHTPyhiAPIrAs1FQR4BXA&dib_tag=se&keywords=keyboard+and+mouse+touchpad+wired&qid=1735002138&sprefix=keyboard+and+mouse+touchpad+wired%2Caps%2C140&sr=8-3
- Antenna for Mesh Device - https://www.amazon.com/915MHz-Antenna-10dBi-Meshtastic-Range/dp/B0DLKTZ66P?dib=eyJ2IjoiMSJ9.y-BOrbqbzywlGMELuU8d6KnJ__jJDcYyUCBLpiDaA3NiXiSyQnIyGy56lgDX_uGxTgr2sGi6W84UxLu9d_OauZKtEy_MUlXEjfAogCz4SsonzGZht1IzQJzYWGPoWL_uGEVoA-

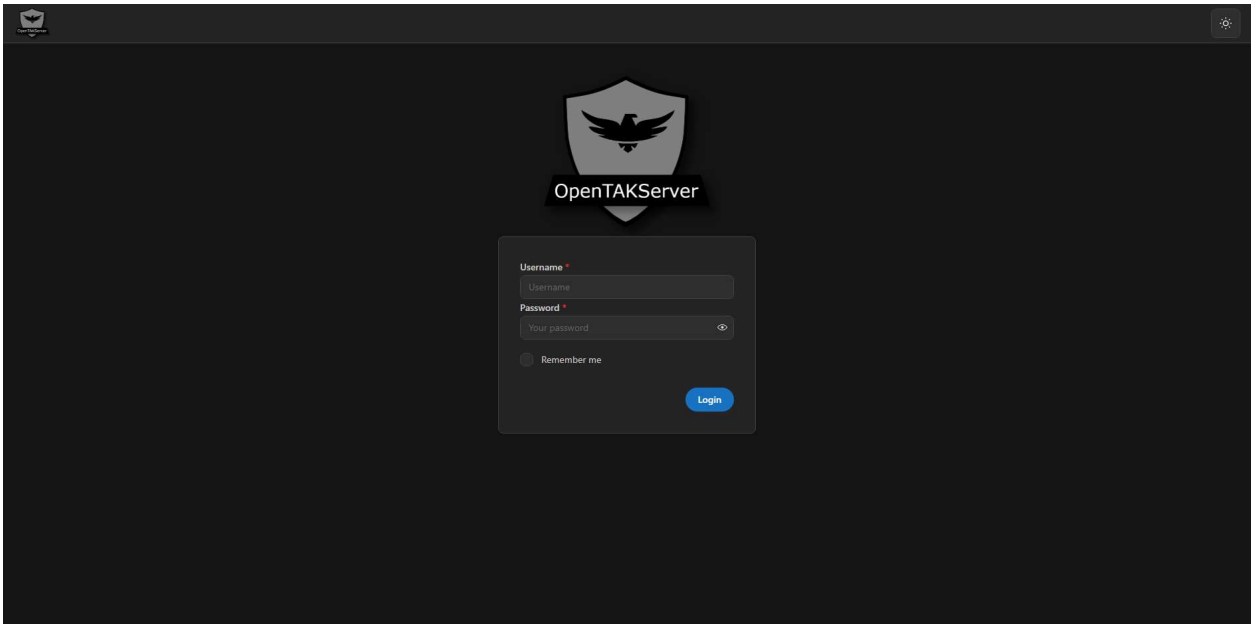
imoSr9RtJj-pFZ5k7SJylzi8niK0ZJKAUlLJNabauvuBV3sevxm21g3oio6uRRGuXSm5qOthAGN4TNZTagjb4VnD1C7JcUTTfGrXU.Z6bH_ip8OmRX5XqJORSohRn0LQrktzqy8cmGx8TfoEs&dib_tag=se&keywords=915mhz%2Bantenna&qid=1735267236&sr=8-5&th=1

Software Needed:

- OpenTakServer - https://docs.opentakserver.io/#
- Ubuntu desktop or Raspberry pi os - https://ubuntu.com/download/desktop or https://www.raspberrypi.com/software/

Setting The Server Up:

- Step 1: Get an image of Ubuntu Desktop or Raspberry pi OS and flash it to the raspberry pi 5.
- Step 2: Installing OpenTakServer on Ubuntu and Raspberry Pi 5, For Ubuntu Run "curl -s -L https://i.opentakserver.io/ubuntu_installer | bash - *" For Raspberry Pi 5 Run "curl -s -L https://i.opentakserver.io/raspberry_pi_installer | bash –".*
- Step 3: Make sure to say No when Prompted, do you want to install "ZeroTier" but make sure to say Yes to everything else.
- Step 4: After OpenTakServer is installed, type your device ip on your browser and it will open up OpenTakServer management WebUI



- Step 5: The Default Credentials are User:administrator Pass:password, make sure to change them ones you login by going to admin then users and click reset password.

- Step 6: After your able to login and reset password, now you will need to go into a terminal and edit the config.yml file by going to “cd ots/” then “nano config.yml” and you will look for a setting call “OTS_ENABLE_MESHTASTIC” it will be set to false and you will set it to true then find “OTS_MESHTASTIC_TOPIC” and it will be set to opentakserver, now change it to something you like, reboot device.

Flashing Meshtastic Device:

- Step 1: Go to " https://meshtastic.org/docs/getting-started/serial-drivers/esp32/" and Download "CP210X USB to UART bridge" and "CH9102 Drivers For Windows".
- NOTE: Important!!!!! -> Make sure to extract both downloads into one folder.
- Step 2: Connect the Meshtastic device and go to "Control Panel\Hardware and Sound\Devices and Printers" look under Unspecified for your device then open device manger and look for it.
- Step 3: After finding your device right click on it and select "Update Driver" then select "Browse my computer for drivers" and click "Browse" and look for the folder with both files extracted to and select the whole folder and click next".
- Step 4: Visit "https://flasher.meshtastic.org" and just follow the steps.

Setting Up Meshtastic Device For Server Ready:

- Step 1: After flashing your Mesh Device, you need to connect it to a phone through Bluetooth, enable your Bluetooth on your device look for Meshtastic device with its own unique name then enter the pin and you will connect.
- Step 2: Install the Meshtastic App through the Play store or Appstore and make sure you are connected and set the region to where you are located.
- Step 3: Ones you’re in the Mestastic App and connected to your device, you will go and click the 3 dots on the top right and go to “radio configuration”.
- Step 4: Now click Device and make sure is set to client, after that go to network and connect to the Wi-Fi that your server is connected to.

- Step 5: For MQTT

COMMUNICATIONS APPLICATION GROUP

Slevin's Guide to Setting up Meshtastic on Open TAK Server

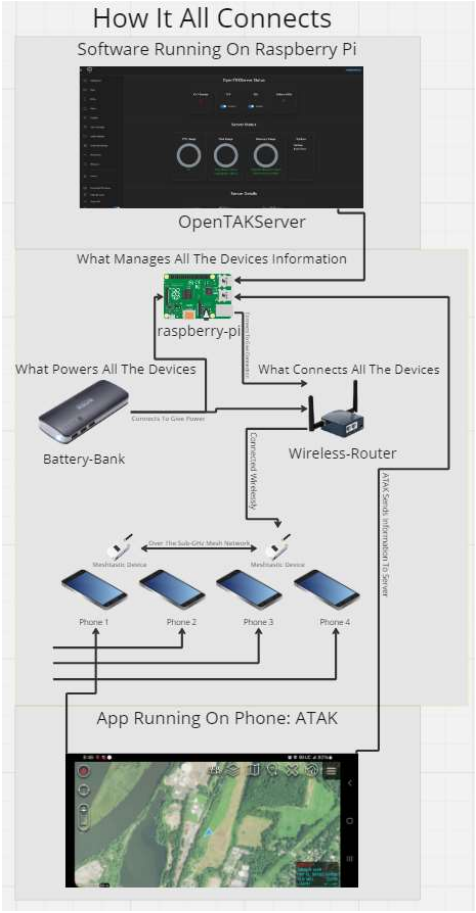
- Edit the config.yml (/home/takadmin/ots/) and verify the following settings are configured:
  - OTS_ENABLE_MESHTASTIC: True
  - OTS_MESHTASTIC_TOPIC: opentakserver
- Setup your Meshtastic "gateway" node:
  - Gateway node must be connected to your network via wi-fi or direct ethernet connection
  - Gateway node's "role" must be set to either CLIENT or ROUTER_CLIENT
  - Enable MQTT on the gateway node:
    - MQTT: Enabled
    - MQTT Server Address: Your server's address
    - MQTT Username: A valid username from an account on your server
    - MQTT Password: The password for the above username
    - Encryption: Disabled (Messages will be encrypted in transit with TLS)
    - TLS: Enabled
    - Root Topic: Value of the OTS_MESHTASTIC_TOPIC setting in the config.yml, which defaults to: opentakserver
    - Proxy to client: Disabled
    - Map Reporting: Enabled (Optional)
    - Map Report Publish Interval: 86400 (Optional)
    - Position Precision: High (Optional)
- Setup Meshtastic channel from the OpenTAKServer's Web UI:
  - Add a channel and make sure that "uplink", "Downlink", "TX" options are set to enabled
  - After adding the channel, use the QR code to add the channel to the Meshtastic gateway node
  - Confirm that the channel name is identical in both the OTS Web UI and on the Meshtastic gateway's channel list
- Troubleshooting:
  - RabbitMQ:
    - Config file(s): /etc/rabbitmq/rabbitmq.conf

Meshtastic Atak Setup:

- Step 1: Go to https://github.com/meshtastic/ATAK-Plugin and download the plugin and after it downloads open it up.
- Step 2: Download Atak App and open it, after its open go to plugins and install the Meshtastic plugin.

- Step 3: Now go to “Settings/Tool Preferences/Specific Tool Preferences/Meshtastic Preferences and Enable “Enable relay to server” and “Enable relay from server”.

Connecting Atak To Server Using SSL:

- Note: Each device that will use "SSL" needs to download the "Truststore Certificate" from the server or be ready with an "USB Flash" or a quick "Python Server".
- Step 1: Go to the OpenTakServer and make an account for each user that will connect to the server for SSl.
- Step 2: Open the Atak app and go to "Settings/Network Preferences/Network Connection Preferences/Manage Server Connections" and click the 3 dots on the top right and click add.
- Step 3: Add any name, add the server address that would be the OpenTakServer, check "Use Authentication" and add the user and password provided by the admin, check "Enroll for Client Certificate" and make sure "Enroll with Preconfigured Trust" is check.
- Step 4: Check "Advanced Options", select "SSL", make sure the "Server Port" is "8089", uncheck "Use default SSL/TLS Certificates".
- Step 5: Click "Import Trust Store" and find the "Truststore Certificate" you downloaded from earlier, and for the password type "atakatak", then click ok.

Server Diagram: